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Quantum-like architecture of the social atom: mental marker coherence and the triple-resonance mechanism of cognitive activation

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Social Laser Theory (SLT) provides an innovative quantum-like mathematical formalism for tracking complex collective social activation dynamics. While its macroscopic manifestations structurally mimic laser physics, the underlying driver stems from the internal state configurations of the individual “social atom.” Broadening this framework, this Perspective article introduces a prescriptive, quantum-like conceptual architecture for the individual cognitive system, formalizing the social atom as a composite entity where mental markers—spanning cognitive, affective, and valence dimensions—are represented as internal degrees of freedom in an abstract Hilbert space. We propose a theoretical framework wherein social activation is modeled through a hypothetical Triple-Resonance Mechanism: the simultaneous quantum-like matching of semantic topic, energetic impact, and valence stance between the agent’s internal state and the informational field. We detail how the recursive processing of quantized informational units (“infons”) can be mathematically represented as a state of cognitive population inversion, and link it to Human Entanglement Theory (HET) by suggesting that the emergence of coherent informational fields is rooted in the phase alignment of these internal markers across a population. By situating SLT within Social Quantum Field Theory (SQFT), this paper offers a neuro-cognitively informed foundation for collective phenomena like emotional contagion and rapid narrative amplification. Our conceptual model suggests that the “Social Laser” should be understood not as a literal physical mapping, but as a flexible analytical manifestation of quantum-like effects across idealized descriptions of human cognitive and neuro-social architecture, serving as an epistemic baseline to stimulate future experimental design and contextuality testing.

KEYWORDS

cognitive and affective markers, quantum-like modeling, social laser theory, social quantum field theory, triple-resonance mechanism

1 Introduction: from neuro-cognitive mental markers to socio-political force fields

The rapid development of quantum information theory—often described as the *second quantum revolution*—has stimulated a profound shift in our understanding of

complex systems. This evolution has sparked renewed interest in the foundational role of quantum theory and its potential applicability across neuroscience and cognitive science. A significant outcome of this progress is the emergence of *quantum-like modeling* (QLM), an interdisciplinary research program that applies the probabilistic and mathematical structure of quantum theory to complex systems in neurobiology, cognition, decision theory, economics, and the social sciences (Khrennikov, 2009; Khrennikova, 2020; Aerts, 2009; Aerts et al., 2013; Atmanspacher, 2004; Atmanspacher and Römer, 2012; Bruza et al., 2015; Busemeyer, 2023; Asano et al., 2011, 2015; Bagarello, 2019; Bagarello et al., 2023; Pothos and Busemeyer, 2022; Plotnitsky, 2023; Plotnitsky and Haven, 2023; Ozawa and Khrennikov, 2019, 2021). This article specifically explores how these quantum-like effects manifest across the human cognitive architecture, from neural signal processing to the higher-order organization of collective mental states.

1.1 The social atom as a neuro-cognitive qubit

Within this broader program, *Social Laser Theory* (SLT) (Khrennikov, 2015, 2016, 2020; Khrennikov et al., 2019; Alodjants et al., 2022, 2023) has been proposed as a robust framework for modeling large-scale social activation phenomena by drawing inspiration from the physics of stimulated emission. However, the efficacy of the “social laser” depends fundamentally on the internal architecture of the *social atom*—the individual information-processing agent (person). In our framework, we move beyond viewing the person as a classical rational actor; instead, we formalize the individual as a composite quantum system. Here, *mental markers*—spanning cognitive, affective, and valence-based dimensions—represent internal degrees of freedom within a complex Hilbert space.

These mental markers function as neuro-cognitive energy levels, reflecting a motivational readiness for participation in collective action. The transition between these levels is triggered by discrete informational excitations called *infons*. Unlike classical stimuli, infons interact with the social atom through a *Triple-Resonance Mechanism*, matching the agent internal mental markers in a process that mirrors the absorption and emission of photons in physical systems. Infons with matching markers lead to an excitation of energy levels related to pro-collective action potentials.

1.2 Parallelity, orthogonality, and neural alignment

The degree of alignment between an agent internal state and the incoming information is captured by the semantic overlap coefficient Γ_{kj} . In quantum-neuroscience terms, this represents the absence of *orthogonality* (opposition) between the agent state and the informational field. When this “social parallelity” is high, the transition probability $P_{0 \rightarrow 1}$ increases, reflecting a neuro-cognitive alignment that facilitates message acceptance. Conversely, as the mental markers of the agent and the infon approach

orthogonality, the probability of rejection, counter-mobilization, and conflict dominates.

Crucially, these cognitive states do not respond to stimuli in a linear causal way. Instead, state-transitions exhibit critical *quantum-like vanishing points* and *acceleration points*, mirroring the developmental dynamics of quantum vortices in physical quantum systems (Aspalter, 2026a,c). This suggests that the brain’s “ready-state” for social mobilization is a non-linear accumulation of mental-marker excitations, eventually leading to a state of *population inversion* within the social group, where the majority of agents are primed for coherent activation.

1.3 Foucauldian fields and human entanglement

To understand the “tuning” of these mental markers, we must look beyond direct signaling. Causality in human action is a complex bridge between direct informational impact and broader *Human Entanglement Theory* (HET), *Mental Entanglement Theory* (MET), and *Social Quantum Field Theory* (SQFT) (Aspalter, 2024a,b, 2026a,c; Asano et al., 2011, 2010; Khrennikov and Yamada, 2025; Khrennikov et al., 2025a,b; Bruza et al., 2015; Asano et al., 2015; Basieva et al., 2019). These entanglements represent probabilistic causal relationships inhabiting the bosonic Fock space ($\mathcal{H}_{field}$) of the social system.¹

We propose that these entanglements function as *socio-political mental force fields*. To define their nature, we draw on *Foucauldian power relations* (Foucault, 1975; Gramsci, 2011; Aspalter, 2024a,b, 2026a,c). In our neuro-cognitive model, power is the “Social Field” that pre-configures the social atom’s internal state. Much like a particle in a field, the agent’s mental markers are tuned by a web of norms, laws, and social expectations—a *Foucauldian Field*—making certain transitions statistically more likely even before a specific infon arrives.

1.4 Signaling and self-emergence in the bosonic space

This field-based perspective implies social causality is a manifestation of varied entanglements, ranging from direct signaling to “self-emergent” forces that travel through both conscious and sub-conscious neural pathways. We distinguish between signaling-based entanglements and non-signaling, self-emergent forces and entanglements where agents follow societal fashion or politics through an internal extrapolation of societal and governmental rules (Freud, 1921; Weber, 2012; Aspalter, 2020).

Furthermore, we identify the role of *atmospheric interpellation*—the causally amplifying influence of dominant public sentiment (Aspalter, 2026a). These collective sentiments act as macroscopic mental force fields that inhabit the bosonic

¹ We also point to the works Eberly et al. (2016), Bardella et al. (2024a,b), Buice and Cowan (2007, 2009), and Gentili (2021) focused on developing field models - both classical and quantum - of neural activity in the brain.

Fock space. As Hayek (1990) emphasized regarding the ideological climate, and Wieser (1926) depicted as the internal powers of the state, these fields are individually anchored yet collectively coherent. Even Pericles noted these intangible forces that penetrate the human psyche, dominating areas of action untouched by direct policy. In the context of this Special Issue, we argue that these forces represent non-signaling-based, self-emergent human entanglements that organize the quantum-like probability distributions of human cognition and collective decision-making.

1.5 Structural analogy between physical and social lasers

In quantum optics, the radiation field produced by a laser operating above threshold is well approximated by a *coherent state* of the electromagnetic field. Such states exhibit classical-like behavior with a well-defined amplitude and phase and are characterized by Poissonian photon statistics. The crucial feature enabling laser amplification is *phase inheritance*: photons generated through stimulated emission adopt the phase of the existing field, thereby reinforcing the collective radiation mode. Basic analogies between physical and social laser are summarized in Table 1.

An analogous mechanism can be proposed for the dynamics of social communication. In SLT, the informational environment is modeled as a quantum information field composed of discrete informational quanta—*infons*. When communication signals become aligned in meaning, emotional tone, and symbolic framing, the resulting informational field may approach a state of partial coherence in which successive messages reinforce one another rather than interfere destructively. Emotional and ideological coherence is the outcome of greater degrees of inter-agential alignment (parallelity), and greater number of repetitions and/or higher intensity (amplitudes) of emotional groundedness.

A key element of this mechanism is the structure of informational quanta themselves. In Section 4.3 we introduce the notion of affective markers, minimal informational units capable of triggering emotional resonance across individuals.

Unlike complex propositional statements, affective markers often carry little semantic content but possess strong emotional valence. Empirical research shows that emotionally charged signals propagate efficiently through social networks and digital media environments, producing cascades of collective attention and behavioral contagion (Hatfield et al., 1993; Centola, 2018; Wu et al., 2011; Lehmann et al., 2012).

From the perspective of SLT, such markers can be interpreted as elementary quanta of social excitation that raise the energy level of social atoms. Repeated exposure to these signals through media systems, advertising campaigns, or political messaging may gradually increase the excitation level of a population, preparing conditions analogous to population inversion in laser physics.

The amplification process itself depends critically on the emergence of collective alignment within the informational field. In physical lasers, phase coherence allows electromagnetic waves to interfere constructively. In sociopolitical communication, a similar role may be played by the alignment of interpretive frames, emotional orientations, and identity markers across a population.

Section 9 of this article explores how such phase alignment relates to well-established mechanisms in sociopolitical theory. These include *emotional contagion* (Hatfield et al., 1993), the *framing processes* through which social movements construct shared interpretations of events (Benford and Snow, 2000), and the role of emotions in mobilizing collective action (Jasper, 2011). When communicative signals resonate with widely shared cognitive and affective structures, they can propagate rapidly through networks of social interaction.

Another important parallel concerns the statistical regimes of the field. In physical lasers, photon statistics change as the system approaches the lasing threshold: incoherent thermal radiation below threshold gives way to partially coherent emission and eventually to a coherent state with Poissonian photon statistics. Similar transitions may occur in the dynamics of public discourse.

When informational signals are weakly correlated, communication resembles a noisy background of independent messages. As feedback mechanisms such as media amplification, social sharing, and algorithmic recommendation strengthen correlations between signals, collective attention may begin to synchronize. In some cases - such as viral advertising campaigns, political mobilizations, or mass protest movements—communication dynamics may approach a regime of coherent informational amplification.

These phenomena have been studied from several complementary perspectives in sociology and political science, including threshold models of collective behavior (Granovetter, 1978), complex contagion theory (Centola, 2018), and research on collective attention in online communication systems (Wu et al., 2011; Lehmann et al., 2012). SLT does not replace these approaches but provides an additional conceptual framework that emphasizes the role of coherence, resonance, and stimulated informational emission in collective dynamics. Central to this framework is the representation of social atoms as complex information-processing agents inhabiting a high-dimensional state space. Unlike simpler models, the multi-marker approach treats the agent’s internal configuration as a tensor product of marker-specific subspaces, where each marker corresponds to a distinct social topic or mental compartment. The interaction between these agents and

TABLE 1 Basic analogy mapping behavioral dimensions of Social Laser Theory onto foundational physical laser systems.

Physical laser	Social laser theory	Interpretation
Atoms	Social atoms (agents)	Information-processing individuals
Photon	Infon	Quantum of social information
Energy levels	Social energy states	Readiness for social action
Population inversion	Social excitation	Large activated population
Coherent EM field	Coherent infon field	Structured discourse environment
Phase coherence	Discursive alignment	Messages reinforce each other
Photon statistics	Infon statistics	Regimes of communication dynamics

the informational environment is modeled via a triple-resonance condition that integrates semantic overlap, energetic matching, and stance tolerance. By embedding these markers into a semantic manifold [or conceptual space (Gärdenfors, 2000)], we account for associative priming and cognitive interference, where information about one topic can partially excite topologically adjacent markers. This interaction is formally described by an interaction Hamiltonian that couples the internal social energy of the atom to its stance, resulting in an asymmetric susceptibility to informational influence depending on the agent's current state.

Despite these parallels, it is important to emphasize that both physical and social lasers represent *idealized theoretical constructs*. Even in physics, coherent states and Poissonian photon statistics are approximations to real systems affected by noise and environmental fluctuations. Social systems are even more heterogeneous and subject to continuous perturbations from competing narratives, institutional structures, and external shocks.

Consequently, the analogies developed in SLT should be understood as conceptual and mathematical modeling tools within the broader framework of QLM rather than literal sociological descriptions. Their value lies in providing a structured mathematical language for describing how aligned informational signals can generate rapid amplification of collective social behavior.

The application of SLT is grounded in a broader foundational perspective that unifies quantum physics and QLM under the “Bild” (image) conception of scientific theories. As explored in Appendix, this approach moves beyond the “straitjacket” of identifying quantum observables with underlying ontic variables. Instead, it posits that both in the subatomic realm and in the macroscopic domains of cognition and sociology, the quantum formalism provides a blurred, approximate image—an echo of a complex reality. By treating the mathematical quantum structure as an epistemic tool rather than a literal ontological description, we can rigorously model the thresholds and leaps of social activation while acknowledging that these processes emerge from a deterministic substrata of internal psychological states and social configurations that function as hidden variables within the QLM paradigm.

This paper is conceptual and aims to bridge SLT, and more generally QLM, with sociopolitical science. It is intended to attract the interest of researchers from both the natural sciences and the humanities. This bridging objective determines the multidisciplinary structure of the paper.

1.6 Incompatible observables and social complementarity

To clarify how quantum-like incompatibility manifests within the social atom, it is necessary to move beyond abstract non-commutativity and provide concrete socio-political examples of complementary variables. In standard classical decision theory, an agent is assumed to possess a pre-existing, joint probability distribution for all possible queries; answering one question simply reveals a pre-existing state without altering the likelihood of subsequent answers. In our Hilbert-space framework, however,

certain mental markers act as incompatible observables. This means that processing or evaluating one dimension inherently perturbs the cognitive system, creating a state change that structurally prevents the simultaneous, sharp definition of the complementary variable within the same underlying subspace.

A primary example of this social complementarity occurs during the simultaneous processing of a policy's *economic utility* (A) and its *ideological/moral valence* (B). Let us consider an agent evaluating a complex legislative reform:

- Observable $\hat{A}$ measures whether the policy yields direct financial or material benefit to the agent's household ($a \in \{+, -\}$).
- Observable $\hat{B}$ measures whether the policy aligns with the agent's deeply held cultural or nationalistic identity ($b \in \{+, -\}$).

Because these operators act within the same cognitive domain and do not commute ($[\hat{A}, \hat{B}] \neq 0$), they cannot share a common basis of eigenstates. If an incoming informational unit (infor) forces the social atom to first evaluate the policy strictly through the economic lens ($\hat{A}$), the state vector is projected onto an eigenstate of $\hat{A}$. This measurement process collapses any pre-existing superposition regarding identity, maximum uncertainty is injected into the ideological dimension ($\hat{B}$), and a subsequent moral evaluation will exhibit a clear question-order effect (see e.g., Pothos et al., 2021; Wang et al., 2013 on the quantum-like modeling of Brexit impact evaluations).

A second, fundamentally distinct example of incompatibility can be found in the trade-off between an agent's *abstract ideological stance* (X) and their commitment to a *concrete, high-cost action protocol* (Y) on the same topic. Let observable $\hat{X}$ represent a broad, low-stakes assessment of systemic support (e.g., “Do you support environmental protection initiatives in principle?”), while observable $\hat{Y}$ represents a highly specific behavioral commitment involving immediate personal loss (e.g., “Will you vote for a specific 20% tax increase on fuel next week?”).

In a classical framework, these variables are governed by a single, global Kolmogorovian sample space where the conditional probability $P(Y|X) = P(X \cap Y)/P(X)$ strictly requires a pre-existing, well-defined joint probability distribution over both parameters. To model a drop in public support as costs increase, a classical utility or contagion framework must assume that the joint probability $P(X \cap Y)$ is preserved and scales monotonically based on objective cost metrics, thereby respecting the Law of Total Probability.

In our framework, however, $\hat{X}$ and $\hat{Y}$ are non-commuting operators ($[\hat{X}, \hat{Y}] \neq 0$) acting within the same underlying state space. Forcing the cognitive system to first resolve the concrete action protocol ($\hat{Y}$) dynamically perturbs the system and shifts the state coordinates within the semantic manifold $\mathcal{M}_{\text{sem}}$. Once an agent is pushed into a definite eigenstate of rejection regarding the immediate high-cost question ($\hat{Y} = -$), their baseline abstract alignment ($\hat{X}$) is retroactively scrambled and updated through state projection. This induces a clear violation of the classical Law of Total Probability, representing a non-commutative cognitive interference that standard joint probability distributions and linear network metrics cannot structurally formalize.

1.7 The indispensability of Hilbert-space representation

A Hilbert-space architecture becomes strictly necessary when modeling cognitive phenomena that violate classical probability constraints, such as contextual dependencies, question-order effects, conceptual compositionality, and the violation of the Sure-Thing Principle.

Classical models can simulate binary endpoints—meaning they can successfully reproduce the final, observed statistical distribution of two-choice outcomes (such as a Yes/No vote or an Accept/Reject decision). By backward-engineering conditional probability parameters or applying standard Markov chains, classical frameworks can fit the static percentages of a final behavioral split. However, they struggle to formalize the non-commutative interference that occurs when multiple affective and cognitive markers are processed simultaneously before that endpoint is reached. Classical models must treat the decision path as a sequence of commutative, sharply defined conditional updates, whereas a Hilbert-space model tracks the continuous trajectory of state superpositions, capturing the complex phase alignments and holistic interference terms that natively drive the cognitive system prior to state collapse.

To clarify this necessity in stricter foundational terms, it is crucial to emphasize that mere non-commutativity or structural context-dependence does not, on its own, mandate a complex Hilbert space representation, as classical models can occasionally simulate order effects using direct classical influences or adaptive conditional frameworks (Basieva et al., 2019; Cervantes and Dzhaferov, 2018). The epistemic boundary where classical probability models fail—and where a Hilbert-space model becomes mathematically indispensable—is the presence of contextuality (Bruza et al., 2023).

Contextuality emerges when a cognitive system possesses a bundle of random variables measured under varying context conditions that cannot be mapped onto a single, global classical probability distribution (Basieva et al., 2019; Bruza et al., 2023). In our framework, this is formalized by treating the multi-marker state as a structurally non-separable tensor product of cognitive and affective subspaces: $\mathcal{H}_j = \mathcal{H}_{c,j} \otimes \mathcal{H}_{a,j}$. When the Triple-Resonance Mechanism processes an incoming informational unit (infor), the update does not follow a classical path of isolated direct influences; instead, it triggers a non-linear, holistic update of the internal state configurations across the semantic manifold $\mathcal{M}_{\text{sem}}$ where the extraction of meaning fundamentally alters the joint distribution of adjacent markers.

Because the phase-alignment of these internal markers across a population forms a coherent state $|\alpha_k\rangle$, any attempt to project these joint multi-variable interactions onto a classical Kolmogorovian probability space results in a structural violation of classical probability bounds, such as the Clauser-Horne-Shimony-Holt (CHSH) or Contextuality-by-Default inequalities (Cervantes and Dzhaferov, 2018). By applying this architecture to SLT, we establish a precise mathematical scaffolding specifically designed to handle these non-decomposable, contextual interferences, providing the exact process theory required for future experimental contextuality testing capable of formally excluding classical alternatives.

1.8 Cognitive neuroscience alignment and boundaries of scope

To rigorously sit within the domain of cognitive neuroscience, we establish explicit conceptual links between the macroscopic “social atom” and underlying micro-level neural dynamics. The mental markers (m) operationalized in our Hilbert space architecture correspond structurally to high-level, stable neuro-cognitive attractors or distributed neuronal assembly configurations within prefrontal-limbic networks. When an agent experiences state transitions via the theoretical Triple-Resonance Mechanism, this formally translates into a modulation of cognitive ready-states, effectively minimizing the metabolic and cognitive threshold required for behavior execution. While this Perspective article does not report direct fMRI, EEG, or experimental neurophysiological measurements, it proposes a prescriptive mathematical scaffolding for future computational modeling of group synchronization and neural alignment dynamics under intense information conditions.

2 Methodology: analogy as a formal epistemic method

Rather than deploying metaphors as purely decorative descriptors, this manuscript utilizes formalized structural analogy as a core cognitive method for mapping structural symmetries across disparate physical and socio-psychological systems (Hofstadter, 2001). In accordance with cognitive science models of conceptual blending, an analogy functions as a foundational mechanism of higher-order cognition, projecting mathematical constraints from a well-characterized base domain (quantum optics) onto a target domain experiencing high multi-variable complexity (sociopolitical communication networks) (Hofstadter, 2001). By establishing explicit mathematical isomorphisms—such as tracking the variance-to-mean relations of informational cascades using the quantum statistical Fano factor—the method provides a prescriptive language to formalize contextual boundaries. This approach explicitly avoids literal physical reductions (ontology) while establishing a predictive mathematical framework (epistemology) to guide future experimental tracking.

3 Core concepts of social laser theory

SLT offers a QL framework for modeling collective social behavior, particularly phenomena such as sudden mobilizations, protests, revolutions, or mass opinion shifts. Drawing inspiration from quantum optics - especially the principles underlying physical lasers—SLT conceptualizes society in terms of discrete agents (social atoms), quantized states of social energy, and coherent collective excitations driven by information fields. The theory is constructed as a mesoscopic model, bridging individual cognition and macro-social dynamics via an abstract formalism grounded in Hilbert space and information theory.

We provide an overview of the SLT methodology, focusing on core concepts rather than the underlying mathematical formalism or the quantum theory of lasers (for a detailed treatment, see Khrennikov, 2015, 2016, 2020; Khrennikov et al., 2019; Alodjants et al., 2022, 2023; Bohm, 1951; Born, 1954). To assist the reader, each section is divided into two parts: a descriptive overview and a brief mathematical formalization. Those seeking a conceptual understanding may bypass the mathematical blocks during their initial reading.

3.1 Social atoms as information-processing agents

In a cognitively grounded interpretation, the social atom is modeled as an information-processing agent situated within a complex social environment. Rather than viewing it as a purely abstract unit, we treat it as a bounded rational cognitive system endowed with memory, attention, and motivational dispositions. To formalize this, we employ *Weberian ideal-typical markers*, allowing us to represent the agent's internal configuration as a tensor product of marker-specific subspaces. This approach acknowledges that while the agent's behavior is shaped by deep-seated representational structures and affective orientations, these markers remain theoretical constructs designed to map the leaps in social activation rather than literal, exhaustive psychological descriptions.

Crucially, the interaction between the informational environment and the individual social atom does not operate via simple mechanical or classical signaling channels. Instead, we frame the internal updates of these Weberian ideal-typical markers through the lens of Bohmian *active information* (Bohm, 1990, 1994). Within this paradigm, the incoming information field acts not as a mechanical force vector transferring momentum, but rather as a non-local catalyst that activates pre-existing meaning structures within the internal configuration of the agent. This processing mode directly contrasts with Shannon's Mathematical Theory of Communication, which focuses on engineering-level signal-to-noise ratios and transmission capacities while abstracting away semantic content. In our framework, information processing is defined entirely by the contextual extraction of meaning, meaning that the "active information" content fundamentally restructures the internal state coordinates of the social atom.

3.1.1 Social energy

One of the central dynamical variables of a social atom is its level of *social energy* E . This energy is not physical but socio-cognitive: it reflects an individual's readiness for socially meaningful action. One may interpret E as an integrated measure of motivational intensity, attentional engagement with socially relevant information, emotional arousal, and commitment to particular frames of interpretation. In this sense, social energy captures the degree to which an individual is prepared to move from passive observation to active participation.

A critical feature of this framework is that the quantization of energy levels is not merely a mathematical convenience, but a formalization of the "thresholds and leaps" observed in empirical social dynamics (Granovetter, 1978). Empirically, the model recognizes that social movements do not grow at a constant, smooth rate; rather, they remain dormant until specific triggers cause a rapid transition, such as the leap from apathy to activism. Drawing on established models of social cascades, we recognize that individuals often remain in a state of stasis until a critical boundary of social influence or internal activation is crossed (Watts, 2002; Schelling, 1971).

Furthermore, psychological research into commitment suggests that internal activation is not a sliding scale but a series of qualitative states (Lewin, 1951). These discontinuous transitions suggest that internal activation must reach specific, quantized levels before a qualitative change in behavior is triggered. Once an agent's motivational intensity reaches a specific threshold, a "quantized" shift occurs, transitioning the agent from passive observation to active mobilization. By applying the language of quantization to these psychological and sociological boundaries, we gain a rigorous method for representing the non-linear, "all-or-nothing" nature of human commitment (Centola, 2018).

It is critical to evaluate these thresholds against established sociological models of network behavior. Classical framework models such as threshold cascades (Granovetter, 1978), complex contagion theory (Centola, 2018), and random-network dynamics (Watts, 2002) successfully capture binary activation curves via empirical macro-variables. The Hilbert-space structure proposed here does not invalidate these models; rather, it aims to subsume them as phase-decoupled baseline limits while explicitly offering an analytical toolkit to address cognitive interference, contextual dependencies, and contextual order effects that classical models do not explicitly formalize.

3.1.2 Social atoms as two-level quantum systems

The state of each social atom i is modeled as a qubit. Mathematically, this is the normalized linear combination of two basic states: $|0\rangle_i$, representing a passive, uninformed, or disengaged agent, and $|1\rangle_i$, representing an excited, informed, or activated agent. In sociopolitical terms, the internal powers of the state and existing Foucauldian power relations provide the bias or pre-set of the social atom; they define the "ground state" ($|0\rangle$) of the population as a baseline of normalization. Transitions to the excited state $|1\rangle$ then represent a departure from this structural baseline toward active social mobilization. The mathematical machinery is based on raising/lowering operators:

$$\sigma_i^+ = |1\rangle_i \langle 0|, \quad \sigma_i^- = |0\rangle_i \langle 1|, \quad \sigma_i^z = |1\rangle_i \langle 1| - |0\rangle_i \langle 0| \quad (1)$$

satisfy the commutation relations:

$$[\sigma_i^+, \sigma_i^-] = \sigma_i^z, \quad (\sigma_i^+)^2 = (\sigma_i^-)^2 = 0. \quad (2)$$

4 Quantum state spaces of social atoms and infons

4.1 Quantum representation of the multi-marker social atom

This framework models a social atom (social agent) as a composite quantum system characterized by N distinct mental markers. These markers represent the internal degrees of freedom of the social atom and interact dynamically with a quantized information field consisting of discrete units of influence, termed infons.

4.1.1 Marker value and dimensionality

For the sake of analytical clarity, we consider dichotomous markers ($m = \pm$). In this binary configuration: $m = +$ represents a positive or supportive attitude toward a specific social object (e.g., the government); $m = -$ represents a negative or oppositional attitude.

While the dichotomous model serves as the foundational “spin-1/2” analog for social behavior, the framework is naturally extensible to multi-valued markers. For instance, a three-level system can be employed to account for a neutral stance: $m = \{-, 0, +\}$ where $m = 0$ explicitly quantifies a state of neutrality or informational indifference. This multi-valued extension is key for mapping empirical scenarios where informational noise, apathy, or structural non-alignment prevents clean dichotomous separation, proving the flexibility of the local state spaces. In the general case, each marker j (where $j = 1, \dots, N$) inhabits a local Hilbert space $\mathcal{H}_j$ whose dimension is determined by the number of possible attitudes associated with that specific marker which is multiplied by the number of social energy levels characterizing a marker. Generally, each marker has its own energy spectrum $E_{0,m} < \dots < E_{n_m,m}$. Again for simplicity we consider the model in which all markers have two energy levels.

4.1.2 Social atom's state space

The Hilbert space of the social atom operating with N dichotomous mental markers having two-level social energy spectra is the tensor product of N marker-specific subspaces:

$$\mathcal{H}_{atom} = \bigotimes_{j=1}^N \mathcal{H}_j, \quad \dim(\mathcal{H}_{atom}) = 4^N \quad (3)$$

Each subspace $\mathcal{H}_j$ corresponds to the j -th mental marker and is spanned by four basis states representing the combination of social energy (passive/active) and marker value (positive/negative):

$$\mathcal{B}_j = \{|E_{0j}, +\rangle, |E_{0j}, -\rangle, |E_{1j}, +\rangle, |E_{1j}, -\rangle\} \quad (4)$$

To effectively ground human and mental entanglement configurations in a quantum-inspired way without relying solely

on macro-social analogies, we formalize the structural non-separability of cognitive and affective dimensions. Let the local marker space $\mathcal{H}_j$ be further decomposed into a tensor product of a cognitive subspace $\mathcal{H}_{c,j}$ and an affective subspace $\mathcal{H}_{a,j}$, such that $\mathcal{H}_j = \mathcal{H}_{c,j} \otimes \mathcal{H}_{a,j}$. A pure quantum-inspired multi-marker state $|\psi_m\rangle$ is non-separable if it cannot be factorized as:

$$|\psi_m\rangle = |\phi_c\rangle \otimes |\chi_a\rangle \quad (5)$$

This mathematical configuration models an intrinsic cognitive-affective entanglement where updating an individual's intellectual frame (c) instantaneously alters their emotional polarization vector (a), establishing a compositional framework for the phase-alignment constraints tracking narrative spreading.

4.1.3 Hamiltonian of a social atom ($\hat{H}_{atom}$)

The internal energy of the atom is the sum of the energies of all its marker constituents. The energy required for a social transition in the j -th marker is the difference between the excited and ground levels: $\Delta E_j = E_{1j} - E_{0j}$.

$$\hat{H}_{atom} = \sum_{j=1}^N \sum_{m \in \{+, -\}} (E_{1j} |E_{1j}, m\rangle \langle E_{1j}, m| + E_{0j} |E_{0j}, m\rangle \langle E_{0j}, m|) \quad (6)$$

Using the operator $\hat{\sigma}_z^{(j,m)} = |E_{1j}, m\rangle \langle E_{1j}, m| - |E_{0j}, m\rangle \langle E_{0j}, m|$, this is expressed as:

$$\hat{H}_{atom} = \sum_{j=1}^k \sum_{m \in \{+, -\}} \frac{\Delta E_j}{2} \hat{\sigma}_z^{(j,m)} + \text{const.} \quad (7)$$

The Equations 1–7 provide the mathematical basics of the model.

4.2 Quantum information field formalism

This formalism describes the interaction between a social agent and a quantized information field, where energy, topic (marker), and stance (value) are treated as independent degrees of freedom.

4.2.1 The single-infon Hilbert space ($\mathcal{H}_{infon}$)

An infon is a quantum of information defined by three independent variables. Its Hilbert space is the tensor product:

$$\mathcal{H}_{infon} = \mathcal{H}_E \otimes \mathcal{H}_{marker} \otimes \mathcal{H}_{value} \quad (8)$$

A basis state $|E, j, v\rangle$ represents an infon with:

- $E \in \mathbb{R}^+$: The Energy (impact/weight of the information).
- $j \in \{1, \dots, N\}$: The Marker Index (the specific social topic).
- $v \in \{+, -\}$: The Marker Value (the stance or attitude).

These states satisfy $\langle E', j', v' | E, j, v \rangle = \delta(E - E') \delta_{jj'} \delta_{vv'}$.

4.2.2 The marker Hilbert space: $\mathcal{H}_{\text{marker}}$

The marker $\mathcal{H}_{\text{marker}}$ is an N -dimensional Hilbert space that defines the “address” in the form of a mental marker of an infon. It represents the N distinct markers or mental compartments that a social agent can possess.

The marker space is isomorphic to $\mathbb{C}^N$: $\mathcal{H}_{\text{marker}} \cong \mathbb{C}^N$. This space is spanned by an orthonormal basis of the eigenstates of the marker operator,

$$\mathcal{B}_{\text{marker}} = \{|j\rangle\}_{j=1}^N \quad (9)$$

where each $|j\rangle$ corresponds to a unique mental marker. The basis states satisfy the standard Kronecker delta orthonormality condition:

$$\langle j' | j \rangle = \delta_{jj'}. \quad (10)$$

To conceptualize the information field as one unbroken network of mutually informing elements, we specify that the field state does not evolve independently of the social context. Drawing again on Bohmian foundations, the infon field state is embedded with an underlying potential field where the total probability distribution matches a wave-like deployment across topologically adjacent coordinates.

4.2.3 Social interpretation

In the social context, $\mathcal{H}_{\text{marker}}$ represents the categorical taxonomy of the agent's interaction with the information field. An infon in a state $|j\rangle$ is “unambiguous” information regarding a single topic (including its emotional coloring). An infon can exist in a superposition $|\phi\rangle = \sum c_j |j\rangle$, representing “interdisciplinary” or “complex” information that addresses multiple mental markers simultaneously.

4.2.3.1 Information routing

Without this space, information would be a global field affecting all mental aspects equally. $\mathcal{H}_{\text{marker}}$ allows for the localization of informational impact.

4.2.4 The Fock space ($\mathcal{H}_{\text{field}}$)

The information environment is a bosonic Fock space constructed from $\mathcal{H}_{\text{infon}}$, allowing for an indefinite number of infons n :

$$\mathcal{H}_{\text{field}} = \bigoplus_{n=0}^{\infty} \text{Sym}(\mathcal{H}_{\text{infon}}^{\otimes n}). \quad (11)$$

The field is governed by operators $\hat{a}^\dagger(E, j, \nu)$ and $\hat{a}(E, j, \nu)$ satisfying:

$$[\hat{a}(E, j, \nu), \hat{a}^\dagger(E', j', \nu')] = \delta(E - E') \delta_{jj'} \delta_{\nu\nu'} \quad (12)$$

We mention some special states. The social vacuum state $|0\rangle$ (here $n = 0$) represents a silent environment devoid of disseminated information. It is the unique state such that $\hat{a}(E, j, \nu)|0\rangle = 0$ for

all (E, j, ν) . A single-infon state $\hat{a}^\dagger(E, j, \nu)|0\rangle$ represents the presence of exactly one unit of information (e.g., a single message) carrying the concrete mental marker endowed with some amount of social energy. High-intensity states $\hat{a}^\dagger(E_n, j_n, \nu_n) \cdots \hat{a}^\dagger(E_1, j_1, \nu_1)|0\rangle$ (large values of n) represent a dense informational atmosphere (e.g., a viral social media campaign or mass propaganda).

The total energy of the information environment (the Field Hamiltonian) is:

$$\hat{H}_{\text{field}} = \sum_{j=1}^N \sum_{\nu \in \{+, -\}} \int E \hat{a}^\dagger(E, j, \nu) \hat{a}(E, j, \nu) dE \quad (13)$$

The Equations 8–13 formalize mathematically quantum information field.

5 Quantum dynamics of the triple-resonant social atom

Large-scale social mobilization exhibits structural similarities to a laser's transition to coherent emission. This process is governed by the Interaction Hamiltonian ($\hat{H}_{\text{int}}$), which is sensitive to the agent's current state. Sociopolitically, this sensitivity is shaped by Foucauldian power relations and the internal powers of the state, which create a background atmospheric interpellation. In our model, these power dynamics are captured by the Lorentzian energy-coupling function ($g_{k,\nu}(E)$), which determines the trigger point for social activation based on an individual's position within these pre-existing force fields.

The interaction between a social atom and the information field is governed by three simultaneous resonance conditions: semantic (marker), energetic, and valence (stance) resonance. Crucially, the internal social energy of the atom is coupled to its stance, creating an asymmetric susceptibility to information.

5.1 Social energy coupling ($g_{k,\nu}(E)$)

The energy matching is enforced by a Lorentzian profile. The transition energy $\Delta E_{k,\nu} = E_{1,k,\nu} - E_{0,k,\nu}$ depends on both the topic k and the stance $\nu \in \{+, -\}$. This allows for modeling agents who are more easily “triggered” or “convinced” in one stance over another:

$$g_{k,\nu}(E) = \frac{V_0 \gamma}{(E - \Delta E_{k,\nu})^2 + \gamma^2} \quad (14)$$

5.2 Semantic overlap (Γ_{kj}) and topological mapping

To move from the discrete description of markers to continuous (which is convenient for modeling of resonance), we embed both the mental markers of the social atom and the infon field into a common semantic manifold $\mathcal{M}_{\text{sem}}$ [cf. conceptual space (Gärdenfors, 2000)]. Each marker k is characterized by a central

coordinate $x_k \in \mathcal{M}_{sem}$, representing its core thematic “anchor,” while each infon j carries a coordinate x_j .

The overlap coefficient Γ_{kj} is defined as a spatial resonance function, typically modeled as a Gaussian kernel:

$$\Gamma_{kj} = \exp\left(-\frac{d(x_k, x_j)^2}{2\sigma^2}\right) \quad (15)$$

where $d(x_k, x_j)$ is the geodesic distance on the semantic manifold (e.g., Euclidean distance in a simplified linear spectrum) and σ represents the thematic width or “cognitive aperture” of the marker. This formulation implies that a social atom does not perceive information in isolation. Instead, an infon of topic j generates a *distribution of excitation* across all markers k that are topologically adjacent. This accounts for associative priming: information about one topic partially excites related markers due to the non-zero tails of the Gaussian resonance, and for marker interference: if two markers k_1 and k_2 are semantically close ($|x_{k1} - x_{k2}| < \sigma$), they may compete for the same infon energy, leading to cognitive dissonance or reinforcement.

5.3 Stance resonance ($\Lambda_{v,v'}$)

As for mental markers, it is convenient to proceed with fuzzy stances described by a continuous stance set, e.g., $v \in [0, 1]$. We define a stance overlap between the marker’s internal value v and the infon’s stance v' . Using a discrete or continuous representation:

$$\Lambda_{v,v'} = \exp\left(-\frac{(v - v')^2}{2\eta^2}\right) \quad (16)$$

where η is the tolerance width. This ensures that the interaction is suppressed if the infon’s stance is too far from the marker’s current state.

5.4 The interaction hamiltonian

The Hamiltonian incorporates the triple-matching condition, where the energy resonance $g_{k,v}(E)$ is now sensitive to the atomic stance:

$$\hat{H}_{int} = \sum_{k,j} \sum_{v,v'} \int \Gamma_{kj} \Lambda_{v,v'} g_{k,v}(E) \left(\hat{\sigma}_+^{(k,v)} \hat{a}(E, j, v') + \hat{\sigma}_-^{(k,v)} \hat{a}^\dagger(E, j, v') \right) dE \quad (17)$$

5.5 Derivation of transition probability

$P_{0 \rightarrow 1}$

To find the probability of a marker k with stance v transitioning due to an infon (E, j, v') , we solve for $c_1(t)$ in the interaction picture:

$$i\hbar \frac{d}{dt} c_1(t) = \Gamma_{kj} \Lambda_{v,v'} g_{k,v}(E) e^{i(\Delta E_{k,v} - E)t/\hbar} \quad (18)$$

Integrating from 0 to t yields the transition probability:

$$P_{0 \rightarrow 1}(t, E, j, v, v') = |\Gamma_{kj}|^2 |\Lambda_{v,v'}|^2 |g_{k,v}(E)|^2 \frac{4 \sin^2\left(\frac{(\Delta E_{k,v} - E)t}{2\hbar}\right)}{(\Delta E_{k,v} - E)^2} \quad (19)$$

The Equations 14–19 are the mathematical basis of the triple-resonance model.

5.6 Determinants of infon—marker coupling

The transition probability was highly affected by the coupling between the infon in the quantum information field and the corresponding marker in the social atom, including the social energy coupling, semantic overlap, and stance resonance (Figure 1).

First, energy refers to the impact or weight of the information. In social learning (Brady et al., 2023a), people tend to learn from individuals to whom others have preferentially attended, learned or deferred, i.e., prestige bias (Chudek et al., 2012). This tendency was further amplified by the algorithms of relevant platforms (Bärthel, 2018).

Second, the semantic overlap between infon in the quantum information field and that of the social agent also has a significant impact on the coupling. To empirically explain how the semantic overlap can have a substantial impact on the effect of an incoming infon on the transition probability, we can compare how the same infon differs in its effect among different quantum information fields. Previous literature (Brady et al., 2023a) has found that participants tend to disproportionately learn from their ingroups (Nowak, 2006). In digital world, the ingroup information was then amplified (Cinelli et al., 2021) by the platform algorithms. The two processes working together make the users produce more ingroup information (Ceylan et al., 2023). Two important groups among others are empirically studied in the literature. The first one is the cultural group (Kitayama and Salvador, 2024). Traditionally, the world was culturally divided into individualistic and collectivistic societies. In the former society, the self was portrayed as an independent entity composed of internal attributes such as personality traits, motives, goals, and attitudes. In the latter society, the self was described as interdependent with others in their ingroups is prevalent. The interdependent self is characterized centrally by relational attributes, such as social roles and status, which define identity. As a result of the cultural difference, people in collectivistic societies tend to have a holistic cognitive style and to have more intuitive or less categorical modes of inference (Talhelm et al., 2014). Two important groups among others are empirically studied in the literature. The first one is the cultural group (Kitayama and Salvador, 2024). Traditionally, the world was culturally divided into individualistic and collectivistic societies. In the former society, the self was portrayed as an independent entity composed of internal attributes such as personality traits, motives, goals, and attitudes. In the latter society, the self was prevalently described as interdependent with other in their ingroups. The interdependent self is characterized centrally by relational attributes, such as social roles and status,

which defines identity. As a result of the cultural difference, peoples in collective society tend to have a holistic cognitive style and to have more intuitive or less categorical modes of inference (Talhelm et al., 2014). The second one is the language group. It is widely believed that how persons think, perceive and understand the world is highly influenced by the language they speak, i.e., the Sapir-Whorf hypothesis (Kay and Kempton, 1984). Previous literature has found that language can play a significant role in structuring, or restructuring, a domain as fundamental as spatial cognition (Lupyan et al., 2020) and color perception (Regier and Kay, 2009).

Third, the moral and emotional values can also have a significant effect on the coupling of the infon and the social agent. Previous literature (Brady et al., 2023a) has found that people tend to pay outsized attention to moralized (Jackson et al., 2019) and to emotionally arousing information, specifically negatively valenced information, especially when such information is relatively rare (Rozin and Royzman, 2001). In digital platforms, these moralized and emotional information were then amplified by the algorithms (Whittaker et al., 2021) of the platforms, which makes users in the platforms produce more moralized and emotional (Brady et al., 2023b) information.

5.7 The multi-Key selection mechanism

The atom's internal state is modified only when the information field satisfies a triple-key resonance:

1. **Topic relevance:** The infon marker x_j must be semantically close to social atom's marker x_k .
2. **Stance tolerance:** The infon stance v' must match the marker stance v within the width η .
3. **Specific energetic resonance:** The infon energy E must match the gap $\Delta E_{k,v}$. Importantly, if $\Delta E_{k,+} \neq \Delta E_{k,-}$, the agent requires different levels of informational impact to change their mind depending on their current stance.

To simplify the model we proceeded with mental markers as integral labels that is without splitting them into cognitive and affective components. The model can be generalized to cognitive-affective markers framework.

6 Affective markers: from social mobilization and viral advertising to social lasing

In political campaigns (especially leading to mass protests), as well as in advertising strategies aimed at generating new needs—such as emerging consumer trends or shifts in lifestyle—communicators often rely primarily on affective techniques that carry minimal semantic load. These techniques make it possible to introduce an idea into mass discourse in the most cost-effective and wide-reaching manner. The central requirement for affective markers is singular: they must evoke strong emotions. The specific type of emotion is of little importance - anger and disgust are just as effective as amusement or laughter.

From a theoretical perspective, such affective markers can be interpreted as semiotic devices functioning predominantly on the level of connotation rather than denotation, see Barthes (1972). Their communicative power resides not in propositional content but in their ability to mobilize emotional resonance, thereby shaping the field of possible meanings available to collective interpretation. Within discourse analysis, their circulation exemplifies what Laclau (2005) describes as the process of affective investment in signifiers, through which empty or floating signifiers acquire mobilizing force by becoming saturated with emotion. This mechanism also resonates with Habermas (1984) concern that strategic communication, when divorced from rational argumentation, risks privileging manipulation over deliberation. In the domain of communication theory, such techniques align with models of low-information, high-impact messaging (McCombs and Shaw, 1972; Entman, 1993), where affective intensity, rather than informational depth, drives agenda-setting, framing, and the saturation of discourse.

6.1 Empirical corroboration of affective virality

Empirical studies on viral advertising confirm and operationalize these theoretical insights. Eckler and Bolls (2011) show that the emotional tone of viral messages significantly influences both attitudes toward the advertisement and intentions to forward it, highlighting the role of affect as a transmission amplifier in digital networks. Dafonte-Gomez (2015) further identifies the core elements that make viral content successful, emphasizing that motivational and emotional triggers - rather than rational appeals - are the primary determinants of sharing behavior (Dafonte-Gomez, 2015). More recently, Rachmad (2025a,b) in two complementary reports for the United Nations Economic and Social Council, explores the dual role of “viral” and “gimmick” effects in shaping behavioral transformation. His findings suggest that gimmicks, understood as low-information but high-salience communicative acts, can function as catalysts for behavioral contagion and discourse alignment. Together, these studies provide empirical grounding for the notion that affective markers operate as minimal yet potent carriers of influence in contemporary communication ecosystems.

When viewed through the lens of SLT, affective markers may be conceptualized as discrete “quanta” of social energy that resonate within a communicative medium. Much like photons in physical lasers, these emotionally charged semiotic units are capable of exciting social atoms (individuals or groups) into higher levels of social energy (typically affective arousal). The repetition and amplification of such markers within media discourse create conditions of coherence, whereby emotional responses become synchronized across large populations. In this respect, the affective markers identified are the minimal yet potent informational carriers that trigger resonance, phase alignment, and ultimately large-scale collective mobilization. Thus, classical theories of framing and agenda-setting can be recast within SLT as mechanisms that structure (align and amplify) the energy

landscape of discourse, determining which emotional excitations reach threshold levels necessary for social coherence and action.

7 Coherent states in physical lasers

7.1 Coherent states of the electromagnetic field

In quantum optics, a coherent state $|\alpha\rangle$, where α is a complex number, is a minimum uncertainty state of the quantized electromagnetic field, defined as the eigenstate of the annihilation operator:

$$a|\alpha\rangle = \alpha|\alpha\rangle. \quad (20)$$

Its Fock state expansion is:

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \quad (21)$$

where $|n\rangle$ denotes the photon number state with n photons.

These states exhibit classical-like behavior with well-defined amplitude and phase, and a Poissonian photon number distribution:

$$P(n) = \frac{|\alpha|^{2n} e^{-|\alpha|^2}}{n!}. \quad (22)$$

7.1.1 Role in lasers

In a laser operating above threshold:

- Stimulated emission leads to exponential amplification of photons in a single cavity mode.
- The emitted photons are in phase with the preexisting field, resulting in phase-coherent light.
- The quantum state of the field is well approximated by a coherent state $|\alpha\rangle$, especially when the average photon number $\langle n \rangle = |\alpha|^2 \gg 1$.

We highlight that the real field in the laser cavity is not exactly a pure coherent state due to quantum noise and decoherence, but it is very well approximated by a statistical mixture centered around $|\alpha\rangle$:

$$\rho = \int P(\alpha) |\alpha\rangle \langle \alpha| d^2\alpha, \quad (23)$$

where $P(\alpha)$ is sharply peaked for a stable laser mode.

8 Coherent states and activation dynamics in social lasers

8.1 Coherent infon fields

In SLT, the quantum information field—*infon field*—plays the role of the electromagnetic field in a physical laser. Each

infon is a quantum of structured, stimulating social information. Each mode of the infon field is characterized by its social energy E and a set of mental markers $m = (m_1, \dots, m_s)$ which may correspond to a specific discourse frame (e.g., anti-authoritarianism, nationalism, etc.).

A coherent state of infons for mode $k = (E, m)$,

$$|\alpha_k\rangle = e^{-|\alpha_k|^2/2} \sum_{n=0}^{\infty} \frac{\alpha_k^n}{\sqrt{n!}} |n\rangle_k, \quad (24)$$

satisfies the eigenvalue equation $b_k|\alpha_k\rangle = \alpha_k|\alpha_k\rangle$. Here, b_k is the annihilation operator for infon mode k .

These coherent infon states:

- Encode highly ordered, low-entropy semantic configurations,
- Support amplification through phase-aligned social activation, and
- Model viral or resonant ideologies, e.g., slogans or narratives spreading coherently across a population.

The Equations 20–24 are basic for the coherent states formalism.

8.2 Population inversion in society

The social analog of atomic population inversion is a population of agents predominantly in the activated state:

$$|\Psi_{\text{soc}}\rangle = |1\rangle^{\otimes N}. \quad (25)$$

The state Equation 25 represents a society ready for collective action or resonance with a dominant ideological signal.

Once a coherent infon probe (e.g., slogan, meme) interacts with this inverted—i.e., excited and receptive—population, it triggers *stimulated activation*, agents emit new infons aligned with the dominant mode, reinforcing the signal. In SLT the infon field state can be approximately described as $|\alpha_k\rangle$.

8.3 Interpretation

Coherent infon states can be understood as modeling structured ideological discourse fields in which meanings are not randomly distributed but organized in a phase-aligned and mutually reinforcing way. Here, the parallelity of aggregate infon states, or infon fields, is translated into collective meaning-creation that give rise to socio-political force fields: such as the construction and amplification of ideological fields, political and socio-cultural atmospheres, increasing polarization of asymmetric power relations, the widening emotional/political gap between rulers (the super-rich and the powerful) and the common people, and so forth.

In this setting, population inversion corresponds to a condition of social readiness for mass mobilization: a situation in which a sufficiently large portion of the population occupies an excited cognitive or emotional state, making large-scale activation possible. The presence of coherence is crucial, because it enables the

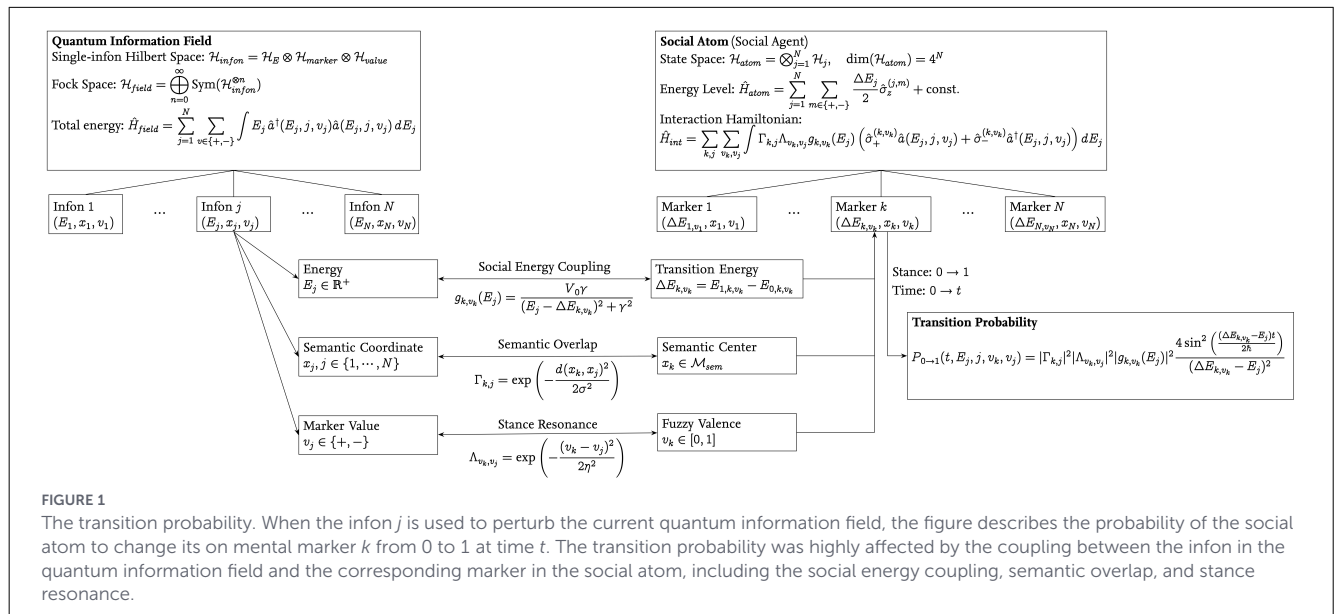


FIGURE 1

The transition probability. When the infon j is used to perturb the current quantum information field, the figure describes the probability of the social atom to change its on mental marker k from 0 to 1 at time t . The transition probability was highly affected by the coupling between the infon in the quantum information field and the corresponding marker in the social atom, including the social energy coupling, semantic overlap, and stance resonance.

exponential amplification of meaning, allowing certain narratives or symbols to grow rapidly in intensity and influence.

Within the SLT framework, the introduction of coherent states provides a powerful conceptual tool for explaining how collective resonance can emerge from communication that would otherwise appear noisy, fragmented, or weakly structured.

In other words, in relational/conformal social quantum field theory terms, angular orientations have been polarized into pro this and contra that, each group/pole in opposing terms to the other. Within the prepared/excited (or activated) population, also called inverse population in laser theory, angular orientations have been changed, relationships altered, meanings, motivations and readiness to act have been amplified and/or diametrically altered (pluses turned into minuses and vice versa). In general, relative degrees of similarities alter the course of politics, economics and the society as a whole (Aspalter, 2026a,b,c).

9 The phase of a coherent state in physical and social lasers

In the definition of a coherent state $\alpha \in \mathbb{C}$ is a complex number. This number encodes both the amplitude and phase of the state $\alpha = |\alpha|e^{i\theta}$, with:

- $|\alpha|$: the amplitude (related to the average photon number),
- θ : the phase of the coherent state.

In quantum optics, the phase θ determines the location of the coherent state in phase space, see formulas 26, 27. The expectation value of the field operator is $\langle \alpha | b | \alpha \rangle = \alpha = |\alpha|e^{i\theta}$. Using the quadrature operators:

$$q = \frac{1}{\sqrt{2}}(b + b^\dagger), \quad p = \frac{1}{\sqrt{2}i}(a - a^\dagger), \quad (26)$$

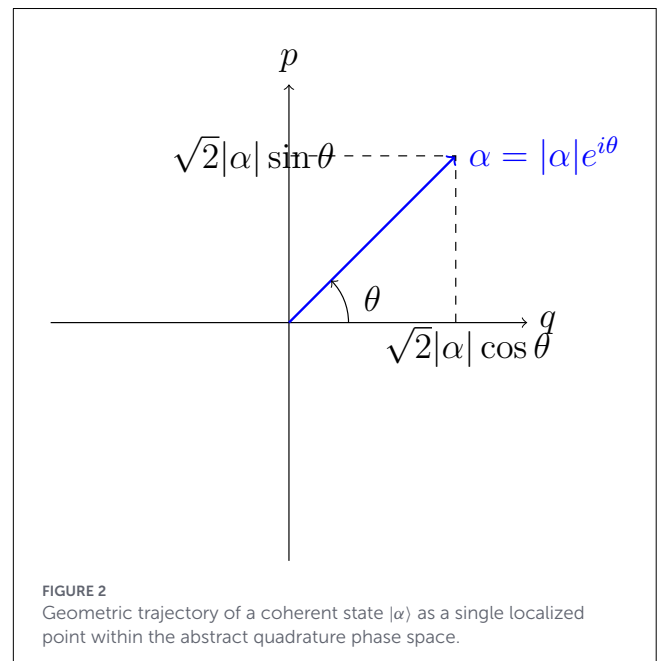


FIGURE 2

Geometric trajectory of a coherent state $|\alpha\rangle$ as a single localized point within the abstract quadrature phase space.

the coherent state's center in phase space is:

$$\langle q \rangle = \sqrt{2}|\alpha| \cos \theta, \quad \langle p \rangle = \sqrt{2}|\alpha| \sin \theta. \quad (27)$$

This structure is illustrated by Figure 2. In laser physics, the phase θ plays a central role in determining how the electromagnetic field behaves under interference. When several coherent beams overlap, it is their relative phases that decide the outcome: aligned phases lead to constructive interference and amplification, while opposite phases produce destructive interference and attenuation. A well-functioning laser emits light that is coherent and characterized by a well-defined phase, although, on longer time scales, unavoidable quantum fluctuations can gradually induce phase diffusion.

The number of quanta (photons) in the coherent state $|\alpha\rangle$ follows a Poisson distribution (Equation 22). This statistical distribution depends only on the magnitude $|\alpha|$, which sets the average photon number, and is independent of the phase θ . Nevertheless, while the phase does not influence the photon-number statistics directly, it remains essential for interference and coherence phenomena, where relative phase differences determine observable effects.

To translate the properties of coherence to the socio-cognitive plane, we recognize that any quantum-like operational environment requires that the most general features of the social atom must be expressed in terms of complementary pairs of variables. Following this systemic constraint, tracking the semantic orientation coordinates of an agent with absolute precision creates an inevitable loss in the structural definition of their affective volatility vector. Consequently, the gradual loss of phase coherence within a target population acts as a formalization of social decoherence, tracking how systemic boundary noise and institutional fluctuations degrade the alignment of macroscopic ideological movements.

In SLT, coherent states of the infon field are used to model structured ideological fields. The phase of a coherent state $|\alpha_k\rangle$ can be interpreted as follows. The phase θ_k represents the alignment of emitted infons with the dominant ideological or political direction. If social agents emit infons with the same phase θ_k , their communication is constructive and reinforcing. Misaligned phases lead to destructive interference, noise, or fragmentation of discourse.

9.1 Collective coherence

The phase coherence of social infons is what makes large-scale amplification possible in the first place. When individual cognitive excitations share a common phase, ideological signals can be reinforced through a mechanism analogous to stimulated emission, so that meanings do not merely circulate but grow in intensity. This shared phase alignment also enables synchronization of collective behavior: individuals respond in a coordinated way, leading to coherent activation rather than scattered, uncorrelated reactions. Under such conditions, certain narratives, symbols, or slogans can rapidly emerge as dominant, since their repetition occurs in phase across the population and thus accumulates constructively.

9.1.1 Summary

The phase θ of a coherent state plays a fundamental role both in quantum optics and in quantum-like models of society. Although it does not influence number statistics directly, it governs coherence, interference, and alignment within the field. In SLT, coherence of phases across the population is a necessary condition for ideological amplification and collective resonance. In this sense, SLT mirrors the functioning of physical lasers, but transposes these mechanisms into a cognitive and discursive space, where alignment of meaning replaces alignment of electromagnetic waves. In the following, the cognitive and discursive space results in the formation of mass social action space.

10 Photon statistics in physical lasers

The way photons are distributed inside a laser cavity depends strongly on how the laser is operating. In fact, the statistical properties of the light change qualitatively as the system moves from weak excitation to fully developed lasing.

When the laser operates *below threshold*, emission is dominated by spontaneous processes. The light behaves essentially like thermal radiation, and the photon number follows a Bose–Einstein distribution. In this regime fluctuations are large: the variance exceeds the mean, $\text{Var}(n) > \mathbb{E}[n]$. Physically, this means photons tend to arrive in bunches, and the field lacks stable phase coherence.

As the system approaches the *threshold*, it enters a transitional regime. Stimulated emission becomes increasingly important, but spontaneous emission and noise are still significant. The photon statistics are no longer purely thermal, yet not fully coherent either. Fluctuations are often enhanced due to strong feedback in the cavity, and the system displays critical-like behavior.

In the idealized picture of a laser operating *above threshold*, the intracavity field approaches a coherent state $|\alpha\rangle$, defined by

$$a|\alpha\rangle = \alpha|\alpha\rangle. \quad (28)$$

In this state the photon number follows a Poisson distribution,

$$P(n) = \frac{|\alpha|^{2n}}{n!} e^{-|\alpha|^2}, \quad (29)$$

with mean photon number $\bar{n} = |\alpha|^2$ and variance $\text{Var}(n) = \bar{n}$. The equality of mean and variance is the statistical signature of ideal coherent light: fluctuations are present, but they are minimal and well-balanced.

Real lasers, however, never perfectly realize this ideal limit described by Equations 28, 29. Even above threshold, several physical mechanisms introduce deviations from exact Poisson statistics. Phase diffusion broadens the spectral line, pump noise injects additional randomness, gain saturation modifies amplification dynamics, and quantum fluctuations or mode competition further perturb the field. Depending on conditions, these effects can produce either sub-Poissonian statistics, where $\text{Var}(n) < \bar{n}$, or super-Poissonian statistics, where $\text{Var}(n) > \bar{n}$. Thus, realistic lasers display a rich variety of fluctuation behaviors.

10.1 Second-order coherence and fluctuation measures

Photon statistics are often characterized through the second-order correlation function at zero delay,

$$g^{(2)}(0) = \frac{\langle n(n-1) \rangle}{\langle n \rangle^2}, \quad (30)$$

which quantifies the tendency of photons to bunch together. For thermal light one finds $g^{(2)}(0) = 2$, indicating strong bunching. For coherent light, $g^{(2)}(0) = 1$, reflecting Poissonian statistics and statistically independent emission events. Values $g^{(2)}(0) < 1$ correspond to antibunching, a genuinely quantum effect that cannot be explained classically.

TABLE 2 Regimes of physical vs. social laser, threshold's dependence.

Regime	Physical laser	Social laser analogy
Below threshold	Thermal (Bose–Einstein)	Weak, random infon emissions
Near threshold	Transition regime	Feedback, mild coherence
Above threshold (ideal)	Coherent state (Poisson)	Aligned infon emission (early coherence)
Above threshold (real)	Deviates from Poisson	Bursts, cascades, viral spreading

Another useful quantity is the Fano factor,

$$F = \frac{\text{Var}(n)}{\mathbb{E}[n]}, \tag{31}$$

which compares the observed variance to that of a Poisson distribution. When $F \approx 1$, the field is close to coherent. If $F > 1$, fluctuations are enhanced, as in thermal or burst-like emission. When $F < 1$, fluctuations are suppressed, signaling nonclassical or squeezed light. The experimental determination of the quantities given by Equations 30, 31 is a complex methodological and technical problem.

10.2 Analogy to social laser theory

In SLT, *infons*—units of information carrying cognitive meaning—play a role analogous to photons. Extending the statistical analogy helps clarify different regimes of social information dynamics, see Table 2.

In a social system operating below threshold, individuals produce and share information in a largely uncoordinated way. Emission of infons is sparse, weakly correlated, and driven mostly by spontaneous personal motivations. The overall pattern resembles thermal radiation: noisy, scattered, and lacking collective alignment.

Near the social threshold, feedback mechanisms such as likes, shares, algorithmic amplification, or media reinforcement begin to synchronize agents. Infons carrying similar cognitive markers become increasingly correlated. Fluctuations grow, and bursts of coordinated communication start to appear. The system hovers between disorder and coherence.

In an idealized above-threshold social regime, a macroscopically coherent state emerges. Large groups of agents emit aligned infons, reinforcing the same narratives, slogans, or symbolic frames. Communication becomes strongly amplified and appears highly coordinated. Statistically, this resembles the Poissonian regime of a physical laser: emission is intense yet structurally ordered.

Real social systems, however, rarely maintain perfect coherence. Differences in attention, competing discourses, external shocks, and media noise introduce irregularities. As a result, communication patterns often exhibit bursts, cascades, and viral spikes—forms of super-Poissonian behavior in which fluctuations exceed the mean. In this way, the analogy with laser physics not only illuminates

the possibility of coherent amplification, but also explains why real social dynamics remain intrinsically noisy and variable.

11 Amplification through phase-aligned social activation: analogies from cognitive and sociopolitical sciences

As discussed in the introduction, one of the central aims of this article is to develop a shared vocabulary that bridges the foundational concepts of SLT with corresponding notions in sociopolitical science. In this section, we explore the relationship between the SLT concept of *amplification through phase-aligned social activation*—a verbal formulation of the coherent-state dynamics characteristic of the idealized social lasing regime—and a range of established ideas from sociopolitical theory, mainly related to resonance, cascade amplification, and social coherence.

We find that many key constructs in political psychology, communication studies, and identity theory can be meaningfully related—at least metaphorically or structurally—to the abstract framework of SLT, and in particular to the coherent state $|\alpha\rangle$ as defined in Equation 24. While these connections are currently indirect and often heuristic in nature, our broader aim is to invite researchers in cognitive and sociopolitical sciences to engage with the QLM paradigm. We anticipate that future work will establish more formal, and ideally mathematical, correspondences between these domains.

This interdisciplinary coupling opens new research avenues, including empirical estimation of social coherence parameters, simulation of social laser effects based on frame diffusion, and analysis of movement success in terms of resonance strength and social energy thresholds.

11.1 Emotional contagion and affective synchronization

Emotions frequently spread across social groups, creating synchronized affective states that enhance collective readiness for coordinated action. A quantum-like model (QLM) emphasizing the dual-track nature of emotional processing—cognitive vs. affective—was proposed in Khrennikov (2021).

Hatfield et al. (1993), in their seminal work *Emotional Contagion*, argue that people often “catch” the emotions of others unconsciously. This occurs through automatic mimicry (copying facial expressions, vocal tones, or postures), feedback loops that reinforce emotions, and physiological coupling. Such mechanisms promote convergence in emotional states, fostering social cohesion, empathy, and coordinated behavior.

11.1.1 Mapping to social laser theory

From the perspective of Social Laser Theory (SLT), automatic mimicry corresponds to phase alignment among social agents,

feedback loops amplify emotional signals, and physiological coupling produces coherence in group affective states. Social cohesion lowers activation thresholds, while emotion propagation mirrors stimulated emission in the social lasing process.

Rimé (2009) extends this perspective, highlighting that emotions naturally drive social sharing. Individuals communicate their experiences almost immediately through verbal expression and repeated narration. Rimé distinguishes primary sharing (direct communication), secondary sharing (listeners retransmitting emotions), and tertiary sharing (broader social diffusion). Empirical studies suggest that 85%–95% of emotional experiences are shared, often engaging listeners emotionally. Cross-cultural research (Rimé et al., 1998) confirms this pattern across Belgium, Turkey, and Japan, with sharing typically peaking within 48 h (Christophe and Rimé, 1997). Experimental studies indicate that sharing enhances cognitive understanding but may not reduce emotional intensity, supporting QLM's dual-track framework (Zech and Rimé, 2005). Secondary and tertiary sharing produce cascading flows of emotional information (Luminet et al., 2000), while large-scale events, such as the death of Princess Diana or the 9/11 attacks, demonstrate mass emotional alignment, described as “collective effervescence.”

In SLT terms, primary and secondary sharing act as stimulated emotional emission, repeated narration pumps emotional energy, widespread sharing aligns phases across individuals, listener resonance strengthens coherence, and propagation cascades trigger large-scale activation.

11.2 Framing theory and discursive alignment

Framing theory explains how social movements craft and communicate messages to mobilize supporters. A frame is an interpretive structure that helps people make sense of events and identities. Movements construct frames by defining problems, attributing causes, proposing solutions, and motivating action, while audiences bring pre-existing frames shaped by culture and experience. Mobilization is most effective when movement frames resonate with audience frames—a phenomenon known as *frame resonance*.

Khrennikov's research on contextual decision-making intersects with framing processes (Aerts et al., 2013; Khrennikova, 2020), and Pothos and Busemeyer (2009) explicitly discuss framing effects in cognitive contexts such as preference reversals.

Snow and Benford (1988) introduced frame resonance, emphasizing that effective framing combines a compelling narrative with deep cultural and emotional connection. This involves diagnostic framing (problem identification), prognostic framing (solution proposal), and motivational framing (rationale for action). Tarrow (1998) stresses that framing is dynamic, shaped by negotiation among movements, opponents, and political institutions.

Recent studies highlight the dynamic and emotional nature of frame alignment:

1. **Dynamic alignment:** Frame resonance evolves with cultural and political context. Benford and Snow (2000) identify bridging, amplification, extension, and transformation mechanisms.

2. **Emotional and moral appeals:** Emotions such as anger, pride, and hope enhance resonance and motivate participation (Jasper, 2011).
3. **Media and digital effects:** Social media accelerates frame diffusion and enables micro-targeted messaging.
4. **Intersectional framing:** Addressing multiple overlapping identities broadens appeal and mobilizes marginalized groups (Cho et al., 2013).

11.2.1 Coupling framing theory with social laser theory

Frame resonance corresponds to coherence in social energy states: just as lasers require phase coherence and population inversion, movements require aligned cultural frames for large-scale action. Discursive alignment synchronizes cognitive states, increasing the likelihood of “social lasing”—rapid, amplified mobilization. Threshold activation in SLT parallels motivational framing, where individuals shift from passive agreement to active participation. Iterative frame alignment mirrors SLT feedback loops, where social energy fluctuates due to internal discourse and external influences, including counter-framing and media dynamics.

Integrating these theories provides a rich toolkit for understanding social movements. Framing offers contextual socio-cultural and narrative grounding, while SLT models amplification and coherence dynamics. This combination enables research on social coherence parameters, simulation of social laser effects from frame diffusion, and analysis of movement success in terms of resonance strength and social energy thresholds (see also article Khrennikova, 2020 for applications to finance).

11.3 Collective attention and synchronization of timelines

In the digital age, collective attention often exhibits rapid, large-scale synchronization across populations, particularly during media events, crises, or breaking news. This synchronization manifests as temporally aligned bursts of online activity—especially on platforms like Twitter—where timelines converge around trending topics. These attention spikes reflect underlying cognitive processes, where millions of individuals orient their perception, emotion, and information processing around a common focus point, effectively creating a shared temporal and semantic frame.

Wu et al. (2011) provide foundational insights into how information diffuses on Twitter by analyzing the “who says what to whom” model of communication. Their study reveals that while most content remains within small networks, high-profile media events or celebrity communications can create bursts of attention that transcend typical network boundaries. This results in sudden synchronization of large numbers of users' timelines, indicating collective attention events.

Building on this, Lehmann et al. (2012) identify distinct *dynamical classes* of collective attention on Twitter. They classify attention patterns into categories such as “sustained interest,” “transient bursts,” and “cyclical reactivations.” Importantly, these

patterns correspond to different kinds of events and topics—for instance, scheduled television events vs. unexpected disasters—each producing a characteristic time-evolution of attention and synchronization among users. The emergence of these classes underscores how collective cognition is temporally structured and modulated by external stimuli, media practices, and social amplification mechanisms.

Together, these works suggest that the synchronization of timelines—observable in tweet volumes, retweet cascades, or hashtag use—serves as a real-time proxy for collective cognitive engagement. Such synchronization may also play a critical role in phenomena like narrative formation, belief propagation, and even protest mobilization, providing a fertile ground for modeling tools like Social Laser Theory or quantum-like attention dynamics.

11.4 Resonance in political communication

Political communication becomes particularly effective when messages *resonate*—that is, align affectively and cognitively with the preexisting concerns, identities, or emotional states of recipients. This idea is central to social identity accounts of populist appeal, such as those developed by Mols and Jetten (2015), who argue that right-wing populist messages gain traction not necessarily due to economic hardship, but because they affirm and activate perceived status threats among strongly identifying majority group members.

To formalize this concept of *resonance* in QLM, we use the formalism of complex Hilbert spaces $\mathcal{H}$, see Equations 32–34. Let $|\psi_i\rangle$ represent the cognitive-emotional state of individual i , encoded as a normalized vector $|\psi_i\rangle$ in $\mathcal{H}$. Let $\hat{M}$ be a Hermitian positively semi-definite operator acting in $\mathcal{H}$ and representing the ideological and affective structure of a political message m , including its symbolic markers, emotional tone, and identity relevance. We define the resonance function $R_i(m)$ as the expectation value of $\hat{M}$ in state ψ_i :

$$R_i(m) = \langle \psi_i | \hat{M} | \psi_i \rangle. \quad (32)$$

This formulation is not present in Mols and Jetten (2015)'s original empirical study, but provides a QL abstraction to capture how certain messages selectively amplify alignment with specific cognitive-emotional configurations. The scalar product reflects the degree of alignment (or “resonance”) between an individual's internal state $|\psi_i\rangle$ and the projected content structure of a message—described by operator M , enabling the modeling of selective susceptibility and communicative impact within populations.

$$\text{Pr}(\text{activation} \mid m) \propto |R_i(m)|^2 \quad (33)$$

In this framework, resonance acts as a filter for message coherence across the population. When many individuals share similar cognitive basis states (e.g., via identity or ideology), high-fidelity resonance can emerge, analogous to phase coherence in wave systems. This can trigger large-scale opinion cascades, polarization, or rapid mobilization.

11.4.1 Integration with social laser theory

Formula (Equation 32) can be used as a starting point toward integration of theoretical and practical studies on resonance in political communication and SLT.

In SLT, resonant messages function as *stimulated emissions*, aligning phase markers across individuals and facilitating coherent social action. If ϕ_i denotes the phase of an individual's ideological alignment and ϕ_m the phase encoded in the message, then the collective coherence can be modeled as:

$$C(t) = \left| \frac{1}{N} \sum_{i=1}^N e^{i(\phi_i - \phi_m)} \right| \quad (34)$$

High $C(t)$ corresponds to macroscopic coherence and readiness for synchronized action, especially if social energy levels have been sufficiently pumped.

11.5 Divergence from classical contagion and threshold models

To understand how the Triple-Resonance Mechanism operates in practice, it is useful to contrast its predictions with those of classical network frameworks, such as Granovetter's threshold cascades or complex contagion models. In standard complex contagion, an agent's transition from a passive to an activated state is treated as a direct function of structural, macroscopic network exposure—essentially counting active neighbors within a specific network topology. While these classical approaches successfully replicate binary macro-activation curves by tweaking parameters *post hoc*, they ultimately reduce the individual agent to an internally static threshold gate.

The Triple-Resonance Mechanism departs fundamentally from this logic by shifting the focus to a Hilbert-space architecture. Rather than tying behavioral shifts to simple exposure counts, this framework introduces distinct, testable dynamics:

- 1. Asymmetric and stance-dependent susceptibility:** An individual's openness to incoming informational units (infons) is not a fixed baseline or historical parameter. Instead, susceptibility behaves as a dynamic variable that shifts non-linearly depending on how well an infon simultaneously aligns with their topic relevance, energetic impact, and stance tolerance. When the transition energy spectra are asymmetric ($\Delta E_{k,+} \neq \Delta E_{k,-}$), the actual informational weight required to trigger mobilization varies dramatically based on the agent's instantaneous polarization vector—an asymmetry that cumulative exposure counts cannot capture.
- 2. Non-linear vanishing and acceleration points:** Classical models imply that increasing exposure monotonically drives up the probability of activation. By contrast, our mechanism maps the agent's internal ready-state through continuous state trajectories and multi-marker phase alignments. Because these internal degrees of freedom are subject to semantic non-commutativity and contextual constraints, the model naturally accounts for sudden cognitive saturation or abrupt narrative amplification. This manifests as critical “vanishing points”

(where massive structural exposure fails to mobilize a population due to phase mismatch) or sudden, explosive “acceleration points” (where a single, well-timed infon triggers a massive cascade because the population has already been primed into a phase-coherent state of population inversion).

Thus, while classical contagion models must alter physical network connections or macro-level node densities to explain unexpected shifts in behavioral spreading, the Triple-Resonance Mechanism demonstrates that rapid, non-linear mobilizations can be driven entirely by internal state contextuality. This provides a clear analytical scaffolding to track discursive amplification as a collective resonance phenomenon, offering a direct strategy to empirically isolate quantum-like cognitive interference from raw structural exposure. Crucially, the formulation presented here serves as a prescriptive theoretical framework intended to guide future, potentially realizable experimental protocols. By articulating these operational boundary conditions, the model sets a rigorous mathematical baseline for prospective empirical verification, inviting targeted empirical testing to systematically validate these quantum-like cognitive predictions against standard network limits.

12 Integrative table of analogies

Integrative physical-social analogies are presented in [Table 3](#).

TABLE 3 Comprehensive overview detailing structural isomorphisms and metrics linking physical laser attributes with macroscopic sociopolitical dynamics.

Social laser concept	Analog	Observable
Phase-alignment of agents	Frame resonance, emotional contagion	Message uptake, protest participation
Stimulated emission	Viral sharing, retweet cascades	Content virality metrics
Threshold crossing	Granovetter thresholds	Participation spikes
Coherent behavior	Flash mobs, mass rituals	Temporal/spatial coordination
Discourse pumping	Media agenda-setting	Repetition frequency, sentiment buildup

13 From conceptual scaffolding further on to empirically informed theory building

In the social sciences (that includes economics and management science) theories are key. They inform and guide the formation and formulation of research questions and research designs, and as a consequence thereof theoretical/mathematical modeling, as well as empirical and experimental modeling.

The new normal in social sciences are not old system theories, but interaction theories, i.e., evolutionary contextual communication-based social (economic, cultural, etc.) interaction theories. We may also understand them as force-based theories, as opposed to boundary- or demarcation-based system theories.

In old/classical social science, Newtonian dynamics are modeled along a completely deterministic approach. Now, with new paradigm of quantum social science on the horizon, we can see that the old entirely deterministic approach/theories are insufficient and obsolete. The 2007/2008 world financial and economic crisis in economics could not be predicted with classical models, as experts and economists using old (neoclassical) theory models were left dumbfounded, out in the cold. Neoclassical theories and models never were close to the pulse and reality of empirical markets, and of course never close to the empirical reality of human action in economics (Mises, 1998). The new dominant rising theory paradigm is quantum social science, and it also includes psychology and all behavioral sciences, i.e. it extends to all life sciences.

With the new quantum paradigm, multitudes of causal forces rule, plus evolution and history of causalities are fully factored in, built on. Context is key, and not fully ignored (as e.g., in neoclassical economics). Social and life sciences are complex sciences, not simple sciences and not simple research topics that can be modeled with and after old Newtonian fully (100 percent) pre-deterministic laws/ways. The new normal and the new rational choice is quantum social science, the old weird choice is Newtonian-based classical social science. The new normal in all the social sciences (and life sciences) is evolutionary and fully contextual, it caters to the reality of non-commutative probabilities of psychological, social, economic, political, and cultural events, series of events and developments.

To make one united, i.e., universal theory, is not easy and takes time. All truly scientific theories have to be universal theories, i.e., universally applicable—otherwise they are not theories in the scientific sense. Scientific theories have to explain all events and developments within certain ranges/classes of events, cases and developments. There are “meta theories” that inform other “grand theories,” and these again may inform and guide the set up and development of macro, meso, and micro theories, or complement the very same. SLT, for example, is a grand theory that bridges both micro theory realms and macro theory ones.

Quantum social science, and with it quantum-like modeling, embodies the much more rational approaches that are now (and have been since the last couple of decades) available to the broad masses of scientists and researchers. The genie is out of the bottle. It cannot be ignored, it cannot be unlearned, and it influences, informs, and leads to new progress in the decades ahead in every more research areas, topics and applications (Aerts et al., 2023; Aerts, 2009; Aspalter, 2020, 2024a).

The social and economic worlds, the cultural and political worlds are ruled by the principles of quantum social science, this is obvious, and this has been proven already by e.g., Aerts et al. (2023) and Orrell (2023), and many others. Aerts et al. (2023) made clear that—universally speaking—quantum-type behavior is valid in human language. Language is the foundation and driving force of every human cognition and decision-making and hence, of course, all human behavior and action. In quantum finance, Orrell (2023) empirically tested and corroborated the supremacy of quantum-like

modeling using large datasets, and with it demonstrated the non-empiricality of traditional classical modeling techniques that time and again failed to predict financial crises.

The empirical evidence proving and/or supporting the supremacy of quantum-like modeling across the board is striking. The relative lack of and/or relatively slow scientific response can only be explained by Thomas Kuhn's "normal science" theory, where out-of-date theories rule, and where new innovative, successfully empirically tested theories are largely ignored for a long time (Kuhn, 1970).

The prime research field in quantum social/life sciences is communication, intrapersonal, and interpersonal communication, and everything that functions as and/or everything can be expressed with communication. This includes all human behaviors (incl. cognition and decision-making) and human actions in general (Aspalter, 2024a, 2025). The smallest units of communications, i.e., words, are the quanta of society, economy and politics. Words, including numbers and symbols, in communication are the "communication quanta" of human cognition and decision-making, human behavior in particular and in general. As Aspalter (2024b) has put it, words in communication are the "least common denominators" that enable "universal quantum gates," without which universal theory-making in the social, economic, political etc. realm would be impossible. Without universal quantum gates, quantum computing mechanisms and quantum-like computing mechanisms in general would be impossible as well. These universal theoretical least common denominators combine all disciplines in the social and life sciences, they unite all theories and research fields. Aspalter (2024b) pointed out that all human-to-machine communications and vice versa, as well as all machine-to-machine communications are also part of the theoretical descriptive and explanatory plane of quantum social science, i.e. geometrically speaking. All of these communications and all the actors of human communications (human themselves, or their creations, like ideologies, ideas, markets, and institutions) can be depicted as points in Hilbert spaces. To be clear, all social, economic and political actors/agents, including social movements and governments, can be depicted as points in Hilbert spaces (Qadir, 1978).

Future directions in quantum social science research are manifold. The demand and opportunities are sheer endless. The development of different types of theories (positive theories including descriptive, explanatory and predictive theories, as well as normative theories) at different levels (meta, macro, meso, micro, or micro-cum-macro) needs all hands on deck. A new research community has been established in the past couple of decades, and is now taking off in terms of theory development, in terms of membership and in terms of modeling techniques and applications (Aerts, 2009; Haven and Khrennikov, 2013; Acharya et al., 2025; Aerts et al., 2013; Khrennikov, 2014, 2020, 2023; Pothos and Busemeyer, 2022; Orrell, 2023).

14 Further research directions

1. **Empirical study:** Use frequency and sentiment analysis of online discourse to track alignment and coherence over time.

2. **Theoretical modeling:** Continue to construct a formal correspondence between concepts of coherence in quantum physics and frame resonance in political communication.
3. **Case study:** Application of SLT to a real-world movement (e.g., Arab Spring, BLM), modeling activation thresholds, discursive pumping, and collective amplification.
4. **Simulation:** Develop agent-based models with oscillatory internal states and simulate collective phase-locking and social cascades.

SLT opens a pathway toward predictive modeling of social mobilization by providing a formal structure to:

- Measure social energy distributions across populations.
- Identify key mental markers driving resonance (first of all affective markers).
- Estimate thresholds for coherence-induced phase transitions.
- Simulate the dynamics of infon injection and collective response.
- Develop further "universally applicable theories" of quantum behavior of people and in policy making, as well as quantum dynamics in and of social systems, markets and institutions (Aerts, 2009; Haven and Khrennikov, 2013).

While still under development, SLT integrates symbolic, emotional, and informational dimensions into a unified model, promising new insights into the nonlinear dynamics of social change.

15 Epistemological limitations and operational commitments

In alignment with the structural framework of the Copenhagen interpretation applied to quantum-like cognition (Busemeyer and Bruza, 2012), we emphasize that programmatic concepts such as "infor," "bosonic Fock space," and "social energy" are deployed here as mathematical epistemic representations rather than ontological physical mappings. Operationally, an *infor* is defined as the minimal discrete semiotic unit capable of perturbing a marker's internal state space (e.g., a standardized linguistic frame or social media token in discourse). We explicitly acknowledge that empirical determination of a complex discursive phase vector presents a substantial measurement challenge. To fully establish the superiority of this architecture over traditional networks, future experimental steps must design explicit contextuality tests, parallel-composition tracking, and inequalities capable of excluding classical alternatives, as underscored by ongoing methodological debates in quantum social analysis (Jaksland, 2023; Tull and Ozawa, 2026).

To sharpen prospective empirical validation workflows without introducing original datasets, we cross-reference our formal mappings with established quantitative templates. For example, the macro-level indicators compiled within the Index of Economic Freedom (Heritage Foundation) provide an ideal dataset for evaluating ruleset coherence and phase variations in social

institutions. Our model's representation of affective and topological non-commutativity can be evaluated against existing linear baselines like the Hierarchical Reasoning Model of Wang et al. (2025). While traditional hierarchical models utilize linear, tokenized multi-attribute decision metrics, cf. Hwang and Yoon (1981), our quantum-like approach formally accounts for semantic context effects and preference interference under uncertainty, drawing on the holonomic perception architectures envisioned by Pribram (1977).

16 Concluding remarks: toward a neuro-cognitive theory of collective resonance

Social Laser Theory (SLT) provides a powerful abstraction for modeling the emergence of large-scale social activation by applying concepts inspired by quantum optics—specifically coherence, excitation thresholds, and stimulated emission—to the human cognitive architecture. By formalizing these analogies, SLT offers a vital bridge between the individual neuro-cognitive state and collective dynamics. This framework highlights that mass mobilization is not merely an aggregation of classical preferences, but a result of phase alignment and the coherent synchronization of mental markers across a population.

The findings presented here suggest that cognitive state transitions, whether at the individual or institutional level, are inherently non-commutative, contextual, and stochastic. Policy evolution and institutional shifts follow a quantum-like trajectory where interference mechanisms—both constructive and destructive—dictate the outcome of decision-making processes. Within this “Social Quantum Field,” *human entanglements* provide the invisible grid that creates, shapes, and upholds collective action. These entanglements facilitate culture and political hegemonies, being communicationally or environmentally entangled (Aspalter, 2024a). Consequently, everything in social science is contextual, mirroring the requirements of quantum mechanics wherever multiple social agents and informational quanta are involved.

By utilizing SLT, sociopolitical and economic case studies can be analyzed through a new lens of “state changes” and “phase changes.” The transition of social energy, driven by informational “infons” and institutional practices, can be translated into predictive models for policy shifts and economic actions (Aspalter, 2025). This allows for a deeper understanding of sudden mass attitude-cum-behavior changes, ranging from public health compliance to shifts in investment patterns and financial market changes.

Crucially, this article has established a foundational coupling between Quantum-Like Modeling (QLM) and key sociopolitical markers such as framing resonance and attention synchronization. While many correspondences remain structural, the theoretical architecture laid out here opens new avenues for transdisciplinary development across the social sciences and economics (Conte et al., 2009; Haven and Khrennikov, 2013; Khrennikov, 2023; Acharya et al., 2025). This shift moves the focus away from classical models of individual rationality (Holcombe,

2018) toward a theory of collective coherence. It invites us to view sociopolitical change not as a simple linear progression, but as a nonlinear resonance phenomenon emerging from the repeated symbolic stimulation of mental markers and shared affective alignment.

Looking ahead, the development of SLT within the realm of neuroscience and cognitive science requires moving from formal scaffolds to empirical validation. We propose several tracks for future transdisciplinary research:

- **Neuro-cognitive simulations:** Designing agent-based systems with oscillatory internal states to explore how phase coherence and infon injection trigger neuro-cognitive cascades;
- **Historical and marker dynamics:** Applying SLT to historical case studies to estimate resonance patterns, while formalizing the dynamics of cognitive and affective markers, emotional valence, and ideological phase as variables influencing infon absorption;
- **Predictive coherence tracking:** Developing indicators to monitor the buildup of coherence in online and physical discourses to anticipate “tipping points” or mass mobilization events;
- **Cross-scale validation:** Estimating activation thresholds and energy buildup in real-world social, economic, and political scenarios to forecast large-scale transformations.

In conclusion, SLT contributes a generalizable modeling scaffold that integrates the symbolic, affective, and neuro-informational components of behavior. By framing social change as a macroscopic manifestation of quantum-like effects across the human cognitive architecture, we open new directions for forecasting the nonlinear dynamics of our increasingly interconnected and reactive global society.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

AK: Writing – original draft, Project administration, Investigation, Supervision, Conceptualization. CA: Methodology, Validation, Conceptualization, Visualization, Writing – review & editing, Formal analysis. LZ: Validation, Formal analysis, Methodology, Conceptualization, Writing – original draft.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. ChatGPT was used to improve style.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fnhum.2026.1882642/full#supplementary-material>

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